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Magnetostratigraphic constraints on the Gondwanan origin of North China: Cambrian/Ordovician boundary results

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SUMMARY

A significant gap in the middle Palaeozoic apparent polar wander (APW) path precludes polarity definition of the early Palaeozoic palaeopole for North China. This problem can in part be resolved by the intercontinental correlation of magnetic polarity patterns across small time intervals. A magnetostratigraphic study was carried out on upper Cambrian to lower Ordovician sediments near Zhaogezhuang (long. 118.5°E, lat. 39.7°N), North China. After stepwise thermal or thermal and alternating field demagnetizations, a characteristic magnetic component with normal and antipodal reversed directions was identified. These data, drawn from 49 samples, yield a north palaeopole at long. 294.6°E, lat. 32.9°N (dp = 3.0°, dm = 5.3°). A concordant magnetic polarity pattern around the Cambrian–Ordovician boundary and lowest Ordovician obtained from different continents favours a Southern Hemisphere origin (∼17°) formation site. Using the Cambrian–Ordovician APW paths between North China and Gondwana, we suggest that the North China block (NCB) was part of Gondwana during the Cambro–lowest Ordovician, and started breaking away from Gondwana in the lower Ordovician. This finding is contrary to some palaeomagnetic models where the NCB was quite separate from Gondwana in the late Proterozoic, and was attached to the 'Pacific' side of Antarctica.

Key words: Cambrian–Ordovician boundary, Gondwana, magnetostratigraphy, North China block (NCB).

1 INTRODUCTION

Asia is perhaps the best place on Earth to study supercontinent formation and its associated processes. It is now well established that east Asia comprises several continental blocks (Tarim, South China, North China, Indochina and Sundaland) that are separated by suture zones. The blocks have not always been in close proximity and at one time ocean floor separated them. East Asia began its assembly in the Permian continuing through to the Cretaceous (Lin *et al.* 1985; Enkin *et al.* 1992). The long-held view is that in the middle Palaeozoic all of the blocks formed part of the northern edge of eastern Gondwana and were quite close to one another (e.g. Metcalfe 1996). However some investigators have examined the blocks' pre-suturing stratigraphic record and structural histories, and suggest that because they appear to have very different Proterozoic–early Phanerozoic histories, the classic Gondwana-origin view is incorrect (Li 1998). Furthermore, it has been suggested that the east Asian blocks formed quite different parts of the late Precambrian Rodinia supercontinent, which began breaking up in the late Proterozoic (e.g. Li 1998; Li & Powell 2001).

The palaeoposition and palaeobiological affinity of the North China block (NCB) relative to Gondwana is an important factor for understanding the global palaeogeographic reconstruction of the continents of the early Palaeozoic. Cocks & Fortey (1990) have recognized the similarities of the early Ordovician faunas (Bathyruid trilobite) and lithologies between North China and Laurentia, suggesting a close connection. On the contrary, Burrett *et al.* (1990) have shown that the close faunal affinities of the stromatoporoids, nautiloids and trilobites between Shan-Thai, North China and Australia during the Cambrian and Ordovician, suggesting that these blocks were in close proximity. This evidence combined with palaeomagnetic data suggest a Gondwana origin for the North China block (Zhao *et al.* 1992; Metcalfe 1996). However, it is clear that any early Palaeozoic palaeobiogeographic reconstruction has to explain why, at certain times, there is a very strong faunal difference between North China, South China and Australia whereas at other times the faunas are similar (Burrett *et al.* 1990).

Attempts to clarify the lower Palaeozoic palaeoposition and movement history of North China are hindered because many rock units have experienced a significant late Mesozoic remagnetization (Zhao *et al.* 1992). The limited number of middle Palaeozoic

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palaeopoles means that the lower Palaeozoic location of the NCB is imperfectly constrained. The objective of this study is to extend the late Cambrian apparent polar wander path (APWP) already obtained from the NCB down to the uppermost Cambrian and lowermost Ordovician, using the typical Cambrian–Ordovician sedimentary section at Tangshan, North China. Using these data in a magnetostratigraphic correlation between the Cambrian–Ordovician boundary and the lowest Ordovician in Australia, Siberia, northern Europe and North China provide constraints on lower Palaeozoic polarity interpretations as well as palaeogeographic reconstructions.

2 GEOLOGICAL SETTING AND SAMPLING

Two key stratigraphic sections (Yushan and Changshan) of the uppermost Cambrian and lowermost Ordovician are located at Zhaogezhuang (long. 118.5°E, lat. 39.7°N), 40 km from Tangshan City in the Hebei province, and about 150 km ENE of Beijing, North China (Zhou *et al.* 1983) (Fig. 1). The sections consist mainly of stable platform sediments in the form of carbonate rocks with monogeneous lithofacies of moderate thickness intercalated with brecciated limestones, shales and clastic rocks.

The top part of upper Cambrian Fengshan Formation comprises grey thin-bedded argillaceous striped limestones intercalated with shales and flat-pebble conglomerates, which yields trilobite fossils such as *Mictosaukia orientalis* (Zhou *et al.* 1983; BGMRH 1989). The overlying lowest Ordovician Yeli Formation consists of shallow marine deposits. Dark grey limestones are intercalated with yellowish green calcareous shales and flat-pebble conglomerates in the lower part, grey thick-bedded limestone interbedded with thin-bedded limestone and flat-pebble conglomerates in the middle part. Medium-bedded limestones interbedded with thin-bedded limestone and flat-pebble limestones conglomerates are found in the upper part. The Yeli formation reaches about 130 m in thickness at both Yushan and Changshan (BGMRH 1989). The sequence grades upward into Liangjiashan Formation, which consists of thick-bedded dolomitic argillaceous limestones interbedded with medium-bedded limestones and dolomites in the lower part, and dominant gray dolomites in the upper part. The lower Ordovician Liangjiashan Formation has a thickness of approximately 160 m in the Changshan section (Zhou *et al.* 1983). The Cambrian–Ordovician boundary is marked by the appearance of the *Dictyonema Flabelliforme* in the Yushan section, and *Cordylodus intermedius* in the Changshan section (Zhou *et al.* 1983). Combined with abundant fossils, such as trilobites, conodonts and graptolites, the Yeli Formation is assigned to the Tremadocian Stage, and the Liangjiashan Formation is equivalent to the Arenigian Stage (An *et al.* 1983; Zhou *et al.* 1983; Yang *et al.* 1986). Approximate absolute ages for the Cambrian–Ordovician boundary and lower Ordovician are taken from Harland *et al.* (1993) and the intercontinental correlations were given by Yang *et al.* (1986).

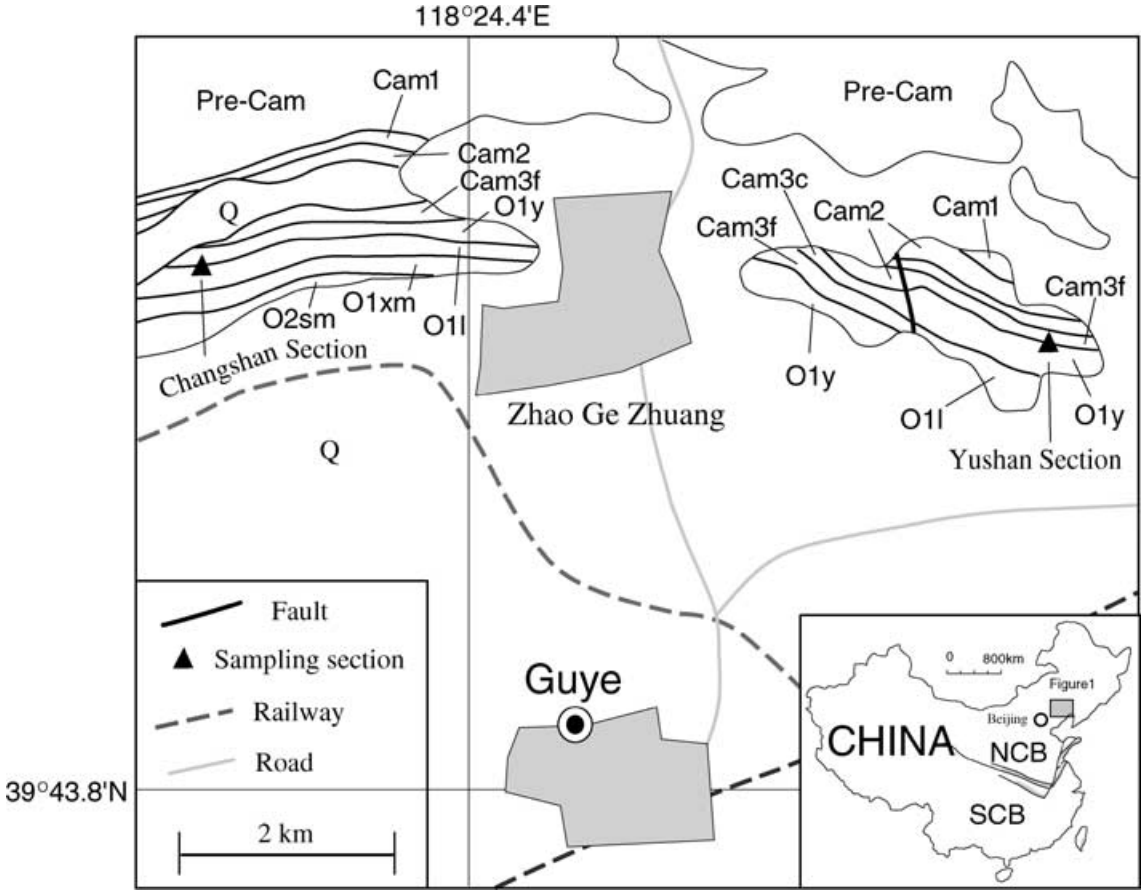


Figure 1. Sketch geological map of study area. Sampling sites are indicated by triangles. Pre-Cam = Precambrian, Cam1 = Early Cambrian, Cam2 = Middle Cambrian, Cam3c = Late Cambrian Changshan Formation, Cam3f = Late Cambrian Fengshan Formation, O1y = Early Ordovician Yeli Formation, O1l = Early Ordovician Liangjiashan Formation, O1xm = Early Ordovician Xiamajiagou Formation, O2sm = Middle Ordovician Shanmajiagou Formation, and Q = Quaternary. Shaded areas indicate the cover of cites.

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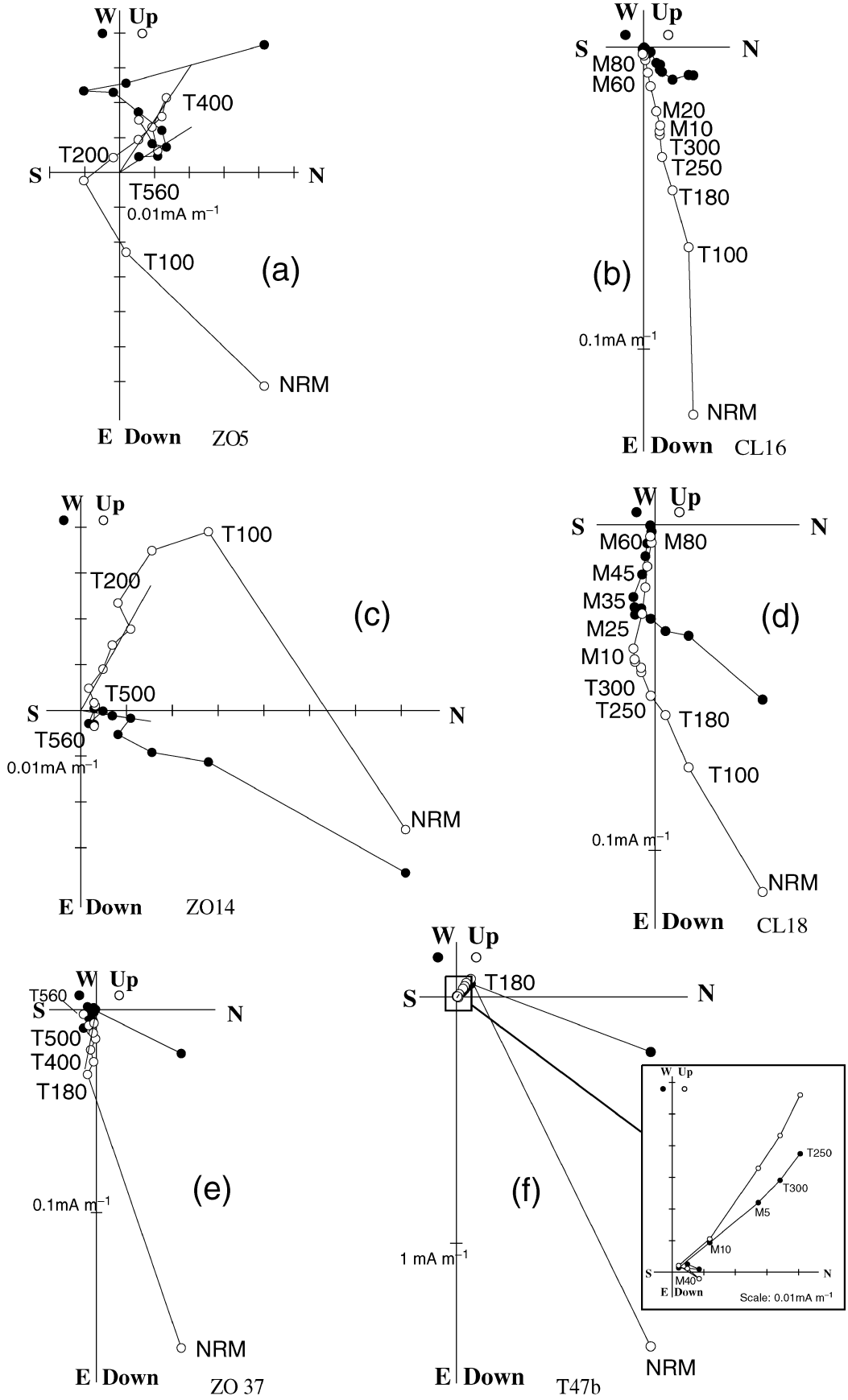


Figure 2. Orthogonal vector projections of thermal and/or mixed thermal and AF demagnetizations of NRM of representative specimens. Solid (open) symbols refer to the projection on the horizontal (vertical) plane in geographic coordinates. T200 = 200°C, M60 = 60 mT.

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The sampled strata dip at about 45° towards the south. The folding probably occurred between the middle and late Jurassic based on the regional studies in the Yanshan area of North China (BGMRH 1989). The conodont reflectance colours demonstrate few signs of metamorphism in Cambrian and Ordovician rocks (An *et al.* 1983, An, pers. comm., 1996). This paper reports first palaeomagnetic results from the Zhaogezhuang sections. 42 oriented drill core samples were collected from the upper Cambrian and the lower Ordovician Yeli formation at Yushan, together with 51 oriented hand samples from thin-bedded argillaceous limestones. Respectively, 23 and 22 oriented cores were sampled from upper part of the Yeli and Liangjiashan Formations in the Changshan Quarry, 4 km west of Zhaogezhuang (Fig. 1). A 0.05–0.2 m sampling interval was attempted within the lower part of the Yeli Formation, and a 1.0 m sampling interval in the upper parts of the Yeli and Liangjiashan Formations.

3 NEW MAGNETOSTRATIGRAPHIC RESULTS

3.1 Treatment

The natural remanent magnetization (NRM) of the samples was measured using the 2G cryogenic magnetometer at the palaeomagnetic laboratory of the University of Kobe. The samples were subjected to progressive thermal and/or alternating field demagnetization, using a laboratory built thermal demagnetizer and a Natsuhara Giken AF demagnetizer. Principal component analysis was used to separate and calculate the different magnetic components (Kirschvink 1980), and site-mean directions were calculated using Fisher's method (1953) or McFadden & McElhinny (1988) statistics.

3.2 Magnetic behaviour

Thermal demagnetization was effectively used in a pilot study. Two magnetic components can be identified on the basis of unblocking temperature ranges (Figs 2a and b): component A (NRM–200°C) and component B (400–560°C). Only a few samples displayed an intermediate temperature component (200–400°C) (Fig. 2a). Some of samples did not yield stable endpoints of the characteristic higher temperature component due to changes in magnetic mineralogy during the heating process. In these cases, we adopted a hybrid demagnetization process which consisted of progressive heating in five steps, to 300°C, followed by AF demagnetization, usually to 100 mT (Figs 2c and d).

Component A has a northerly declination and a steep positive inclination. The mean direction (Dec = 1.3°, Inc = 57.3°, $\alpha 95$ = 3.4°, N = 131) is close to the present dipole field which is Inc = 59.0°, and is probably a viscous remanent magnetization (Fig. 3a). After the removal of the component A, a third of the samples reveal a higher coercivity component which is carried as both normal and reversed polarities (Figs 2 and 3b). Eight samples did not yield a stable endpoint, however, the polarity was usually discernible following using remagnetization circle trajectories. These remagnetization circles were combined with direct directions observed by principal component analysis (Kirschvink 1980) to calculate mean directions (McFadden & McElhinny 1988). The mean direction of component B is Dec = 182.9°, Inc = 32.2° ($\alpha 95$ = 4.7°) in stratigraphic coordinates (Table 1), which corresponds to a North pole position at long. 294.6°E, lat. 32.9°N (dp = 3°, dm = 5.3°).

The mean normal and reversed directions are approximately antiparallel with an angular difference of 167°. Fig. 3(b) shows the northerly upward and antipodal southerly downward directions before and after tilt corrections. The circles of confidence overlap at the 95 per cent confidence level when the reversed direction is inverted, suggesting a single directional population and a positive reversal test (γ_o = 13.0° < γ_c = 14.2°, McFadden & McElhinny 1990). Although a full fold test is not possible, due to strata having monoclinal structure, the characteristic directions differ significantly from Jurassic–Cretaceous and Permian–Triassic directions both before and after tilt correction (Fig. 3b). We therefore consider that component B represents the primary direction.

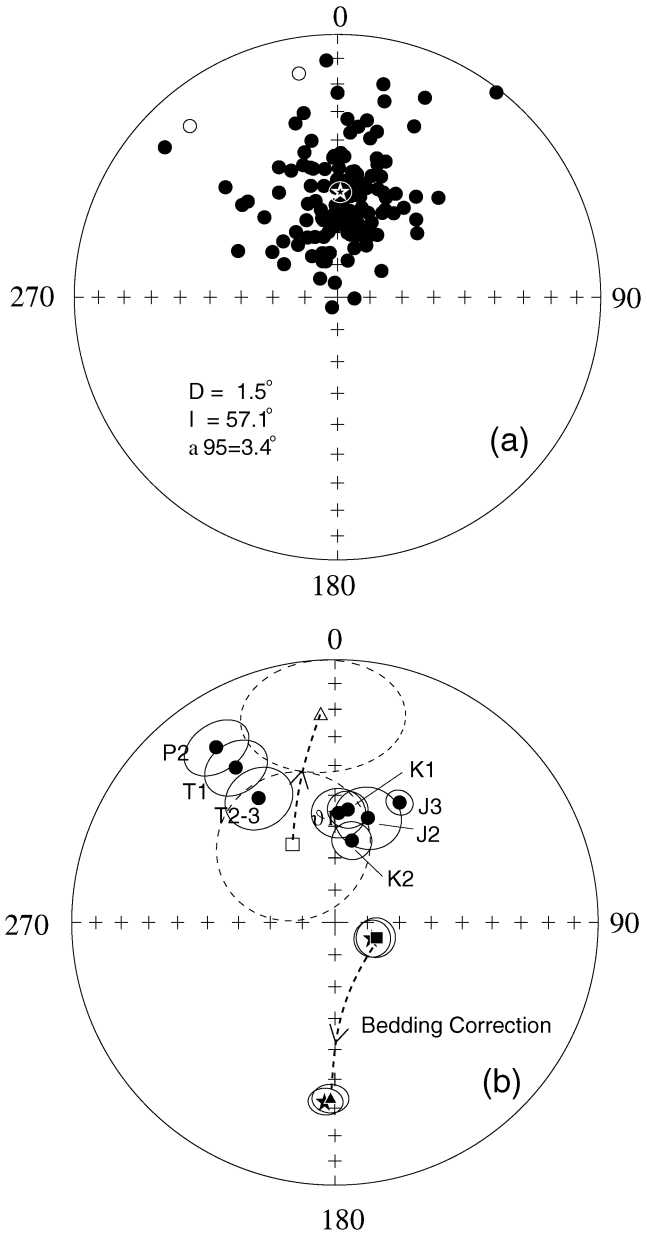


Figure 3. (a) Equal-area stereographic projection of lower temperature components in geographic coordinate. Lower (upper) hemisphere directions are marked with closed (open) symbols. (b) Normal and reversed mean directions are compared with Mesozoic directions deduced from Apparent Polar Wander Path of the North China Block (Enkin *et al.* 1992). Squares (triangles) show means in geographic (tilt corrected) coordinates. The dashed line between the *in situ* and tilt corrected directions show the small circle of tilt corrections.

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Table 1. Lowest Ordovician-characteristic component.

Section (age)	N	Dg (°)	Ig (°)	Ds (°)	Is (°)	α_{95} (°)	k
Yushan (O1)	21	158.9	70.0	183.8	28.1	8.1	16.5
Changshan (Cam/O)	28	69.8	75.1	182.4	35.2	5.7	23.9
Reversed	41	110.6	76.5	181.4	33.3	5.0	20.9
Normal	8	331.4	−62.4	356.7	−21.1	21.1	7.9
Mean	49	113.6	77.4	182.9	32.2	4.7	19.6

Note: N, number of samples; O1, Early Ordovician; Cam/O, Cambro-Ordovician boundary; D, declination; I, inclination (g and s refer to geographic and stratigraphic coordinates, respectively); α_{95}, radius of 95 per cent probability cone for the mean-directions; k, precision parameter.

Acquisition of isothermal remanent magnetization (IRM) is used to identify the mineralogy of the magnetic carriers within the rocks (Fig. 4). The specimens are generally saturated at 0.3 T, which indicates that a low-coercivity mineral is the dominant magnetic phase. The IRM curves increase only slightly above 0.3 T suggesting that a high-coercivity mineral is also present in the carbonate rock. These observations are confirmed by whole rock thermomagnetic experiments. Goethite and magnetite are suggested by Curie temperatures of around 150° and 580°C respectively (Fig. 5a).

Low-temperature IRM experiments demonstrate a spectrum of magnetite grain sizes in the carbonate rocks. Low-temperature IRM of Sample ZO17 is characterized by a rapid magnetization loss at 20 K and a slight decrease around 130 K (Verwey transition) during warming from liquid nitrogen to room temperature using a Quantum Design Magnetic Property Measurement System (MPMS) (Fig. 5b). This indicates that the specimen is dominated by superparamagnetic grains, with pseudosingle domain (PSD) magnetite representing a minor fraction. However, no Verwey transition is observed from Sample CL21 suggesting that magnetite (identified from IRM acquisition tests) resides in the single domain phase. Hysteresis loops measured from most of samples display 'wasp-waisted' behaviour (Fig. 5c), which indicates a bimodal distribution of grains with different coercivities. The wasp-waistedness of these carbonate specimens most likely represents a mixture of superpara- and single-domain magnetites. A logarithmic plot of coercivity (H_{cr}/H_c) and remanence (M_{rs}/M_s) ratios is consistent with a synthetic mixture of single domain plus multidomain grains or superparamagnetic and single domain magnetite assemblage (Fig. 5d), but is distinct from the results reported for remagnetized carbonate rocks (Channell & McCabe 1994).

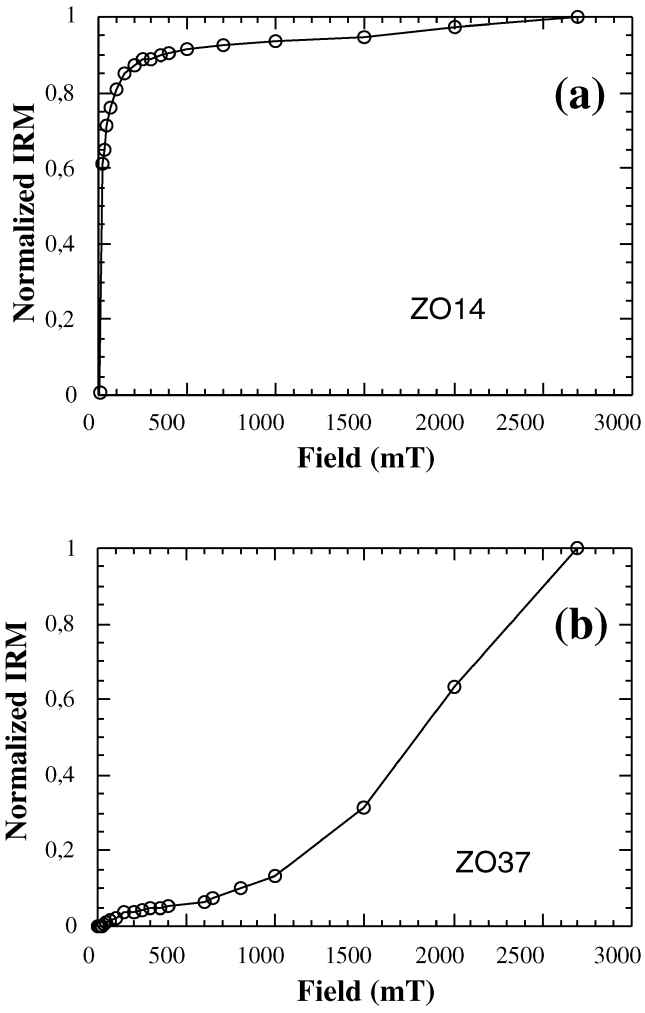


Figure 4. Normalized isothermal remanent magnetization acquisition curves of limestone samples.

3.3 Magnetostratigraphic results

A magnetic polarity sequence for the Cambrian–Ordovician boundary and the lowest Ordovician, based upon the high temperature components of samples examined in this study is shown in Fig. 6. With the exception of four magnetic polarity subzonations defined by a single sample, the magnetozones are constrained by several samples (McElhinny 1978). Magnetic polarity patterns from this study suggest that a predominantly normal polarity zone with several thin reversed intervals occurs at the Cambrian–Ordovician boundary throughout the early Tremadoc Stage. The late Tremadoc and early Arenig is dominated by reversed polarity with several thin normal polarity zones (Fig. 6).

4 DISCUSSION

Recently a set of Cambrian and Ordovician poles was obtained from the Ordos Basin in the western NCB (Huang *et al.* 1999). Results included positive fold and reversal tests. The results of Huang *et al.* (1999) are consistent with the Cambrian–Ordovician palaeopoles previously obtained from different areas in North China (Zhao *et al.* 1992; Yang *et al.* 1997). A refined early Palaeozoic APW path for the NCB has been established at the stage level (Huang *et al.* 1999).

With our new results we extend the Cambrian APW path to the uppermost Cambrian and lowermost Ordovician. Because of a significant gap in the APW path during the middle Palaeozoic, polarity definition of early Palaeozoic palaeopoles is difficult for North China (Zhao *et al.* 1992, 1996; Huang *et al.* 1999). However, this problem can be resolved by the intercontinental correlation of the magnetic polarity pattern around the upper Cambrian–lower Ordovician. In the last decade, lower Ordovician magnetostratigraphic results have been obtained from Australia, southern Sweden and Siberia (Torsvik & Trench 1991; Ripperdan & Kirschvink 1992; Ripperdan *et al.* 1993; Torsvik *et al.* 1995; Gallet & Pavlov 1996; Pavlov & Gallet 1998; Smethurst *et al.* 1998) (Fig. 6). These results suggest that reversed polarity dominated during the Arenigian and Llanvirnian Stages. However, the Cambrian–Ordovician boundary was predominantly normal polarity with short duration reversed polarity intervals as determined in studies from the Black Mountain of Australia, the Kulumbe river section of northwestern Siberia and Dayangcha section of Northeastern China (Ripperdan *et al.* 1993; Gallet & Pavlov

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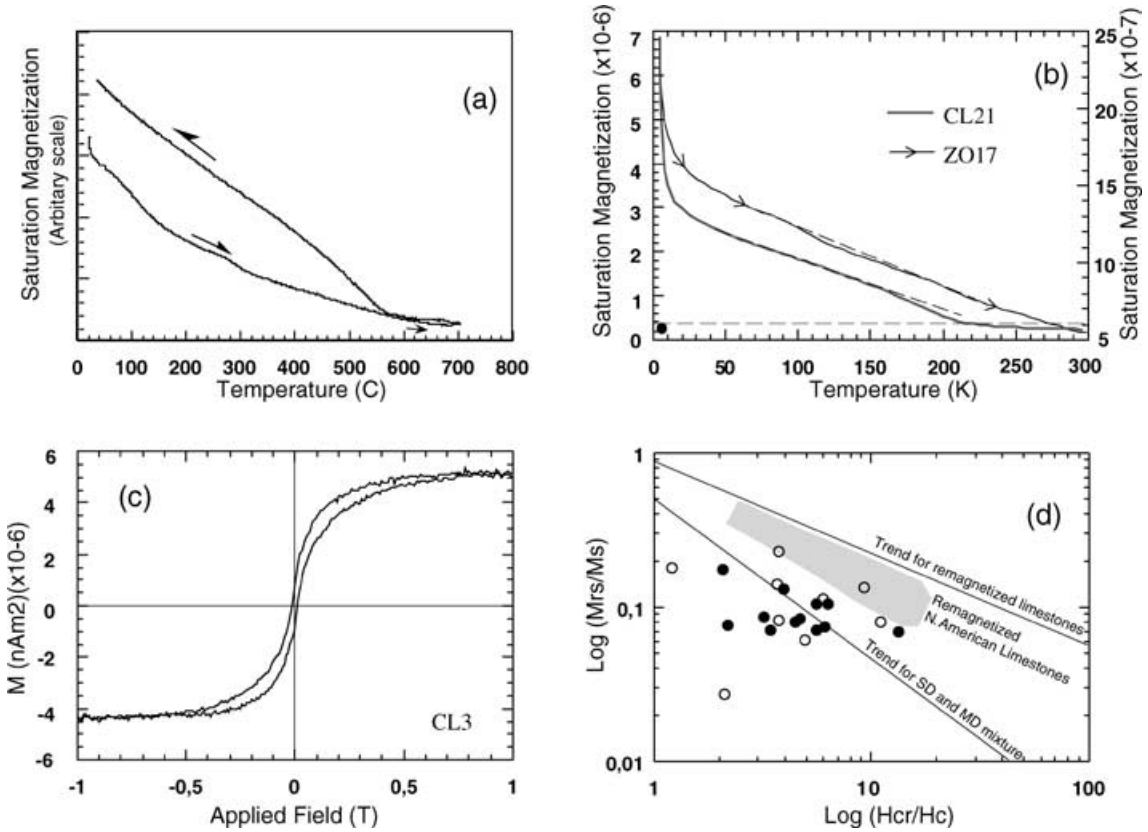


Figure 5. (a) Thermomagnetic curve of a bulk specimen of the lowest Ordovician carbonate rock showing the Curie points of goethite around 150°C and that of magnetite at 580°C; (b) Saturation isothermal remanence (SIRM) of small chips of carbonate rock measured on MPMS during warming from 5 K to 300 K. SIRM was given in a 5000 mT field at 5 K; (c) Hysteresis loops measured at room temperature using a micromag 2900; (d) logarithmic hysteresis parameters plot, trends for remagnetization limestones and for SD and MD mixture are also shown. Dots and open circles represent the normal and reversed polarities specimens.

1996). To a first order approximation, the magnetostratigraphic results from Zhaogezhuang are in agreement with these studies (Fig. 6), e.g. a distinctive normal polarity magnetochron is recorded during the *Cordyolotus proavis* conodont zone in Siberia, Australia and North China, and a predominant reversed polarity is evident during the late Tremadoc and early Arenig. This resolves the early Palaeozoic magnetic polarity ambiguity for the North China Block. We therefore designate our Cambrian–Ordovician boundary and lowest Ordovician pole as a north pole located at long. 294.6°E, lat. 32.9°N ($dp = 3.0°$, $dm = 5.3°$).

For the NCB, the angular distance between the lower Cambrian and lowest Ordovician poles is 46°, which coincides with that of Gondwana (Meert & Van der Voo 1997). It seems contrary to the proposition of a discrete burst of True Polar Wander of up to 90° that occurred during short intervals (15–20 Ma) in the lower–middle Cambrian (Kirschvink *et al.* 1997). However, it is clear that rapid APW (∼30°) occurred during the latest Cambrian and earliest Ordovician, as demonstrated on the major continents, e.g. Laurentia, Gondwanan Australia, Siberia and Baltica (Kirschvink *et al.* 1997). The similarity of the major continent APWP favours the true polar wander hypothesis, and would certainly be important for reconstruction of the palaeopositions of the palaeocontinents.

The earlier reconstructions of Metcalfe (1996), Zhao *et al.* (1996) and Li (1998) that placed the NCB in the northern hemisphere during Cambrian and Ordovician time, were based on the ambiguous polarity choice of early Palaeozoic poles (Lin *et al.* 1985; Zhao *et al.* 1992). Magnetostratigraphic correlation between Australia, North Europe and Siberia during Cambrian–Ordovician and lowest Ordovician time resolves the early Palaeozoic magnetic polarity uncertainty for the NCB, suggesting a southern hemisphere palaeolatitude (∼17°) for the NCB during the Cambrian–Ordovician and the lowest Ordovician (Fig. 7).

The palaeogeographic relationship between the NCB and Gondwana during the early Palaeozoic is a difficult question. There is a loose connection between the NCB and Gondwana in Zhao *et al.* (1996, Figs 3 and 4) and Li (1998, Fig. 5a) has placed the NCB at a low palaeolatitude in the Northern Hemisphere, where the NCB is close to the eastern Australia margin of Gondwana during the middle Cambrian. However, a discrepancy reconstruction placing the NCB in the southern Siberia and Laurentia cratons during the mid-Palaeoproterozoic and latest Neoproterozoic was also proposed by Li (1998, Fig. 3c). Considering the palaeoposition of the NCB in the Rodinia supercontinent of Li (1998, Fig. 3b) these reconstructions require that the NCB and Laurentia separated from East Gondwana around 730 Ma. The NCB then broke away from Siberia and Laurentia at low paleolatitude in the southern hemisphere in the latest Proterozoic. It finally drifted again towards East Gondwana at low palaeolatiutude in the northern hemisphere in the Cambrian and Ordovician (Li 1998, Figs 3c and 5a).

In this study, the Cambrian–Ordovician poles of North China are rotated into the Gondwana reference frame with the NCB sited against southwestern Antarctica. This location and configuration of the NCB is consistent with palaeobiogeographic evidence (Burrett

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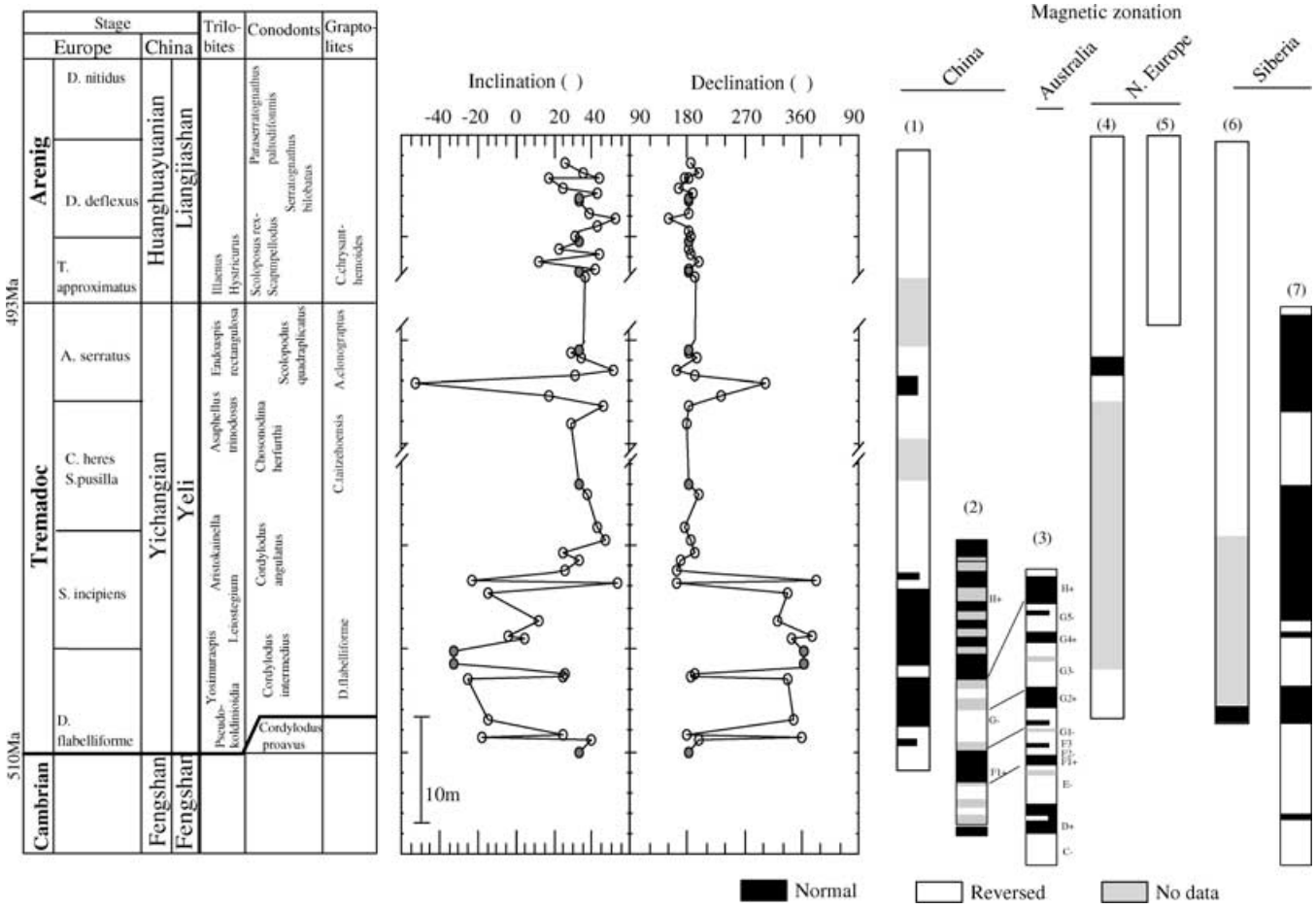


Figure 6. Magnetostratigraphic summary of results from the Zhaogezhuang sections in Tangshan, North China. Grey dots in the inclination and declination columns are values estimated by remagnetization great circles. The polarity defined by a single sample is marked with a small bar. The magnetic polarity patterns from (1) Tangshan of North China, (2) Dayangcha section of Northeast China (Ripperdan *et al.* 1993), (3) Black Mountain section of Australia (Ripperdan & Kirschvink 1992), (4) North Eurasia (Torsvik & Trench 1991; Torsvik *et al.* 1995), (5) St. Petersburg (Smethurst *et al.* 1998), (6) Moyero (Gallet & Vladimir 1996) and (7) Kulumbe rivers sections of Siberia (Pavlov & Gallet 1998) are shown for comparison. The timescale is after Harland *et al.* (1989).

et al. 1990). The connection of the NCB with Antarctica leads to a continuation of the Grenvillian orogenic belt along the eastern margin of Gondwana, and critically avoids the complex movement of the NCB during the latest Proterozoic and early Cambrian (Li 1998). The late Mesoproterozoic active plate margin consisted of a volcanic arc and backarc basin along the southern margins of the NCB (Kuanping complex in the northern Qinling fold belt). The Sm–Nd ages of the Kuanping metamorphic complex range between 1142 ± 18 Ma and 986 ± 169 Ma (Zhang & Zhang 1995). A post-orogenic magmatic intrusion was also dated as 978 ± 79 Ma (Zhang & Zhang 1995). The northern margin of the NCB also displayed a late Mesoproterozoic orogenic belt in the Bainaimiao and Wenduermiao areas, where metamorphism was dated between 961 ± 66 Ma and 1151 ± 76 Ma (Zhang & Wu 1998). These results suggest that both the northern and southern margins of the NCB suffered from the Grenvillian orogenic event. Our early Palaeozoic reconstruction of the NCB and Gondwana also favours an alternative Rodinia model, namely the Australia–Western US Connection (Karlstrom *et al.* 1999; Burrett & Berry 2000), in which the northern Qinling fold belt (Kuanping complex) of the NCB was the western continuation of the Grenvillian belt in the eastern margin of Laurentia (Fig. 7). The reconstruction also shows that the latest Neoproterozoic glacial unit of the Luoquan Formation found along the southern margin of the NCB may be correlated with the Marinoan glacial units in Australia.

Independently, Smith *et al.* (1997) have suggested that the Central Qilian terrane, in the southwestern part of the NCB, can be correlated to the Transantarctic Mountain Fold belt of Gondwana, based on isotope systematics of metasediments in the North Qilian belt of China. The lowest Ordovician pole marks a distinct APW loop for the NCB. The APW paths for Gondwana and North China are separate for this time, which may imply divergence between these two continents (Fig. 7). The close affinity of the early Ordovician tropical bathyurid trilobites and conodonts in the NCB, Laurentia and Siberia, but not in Australia (An *et al.* 1983; Cocks & Fortey 1990), seems consistent with this observation.

5 CONCLUSIONS

Magnetostratigraphic data has been reported from the Cambro–Ordovician of the North China block. The section mean direction determined from the higher temperature magnetic component yields a late Cambrian–early Ordovician north pole at long 294.6°E, lat 32.9°N (dp = 3°, dm = 5.3°). Samples yield a positive reversal test and correlate with intercontinental magnetic stratigraphy

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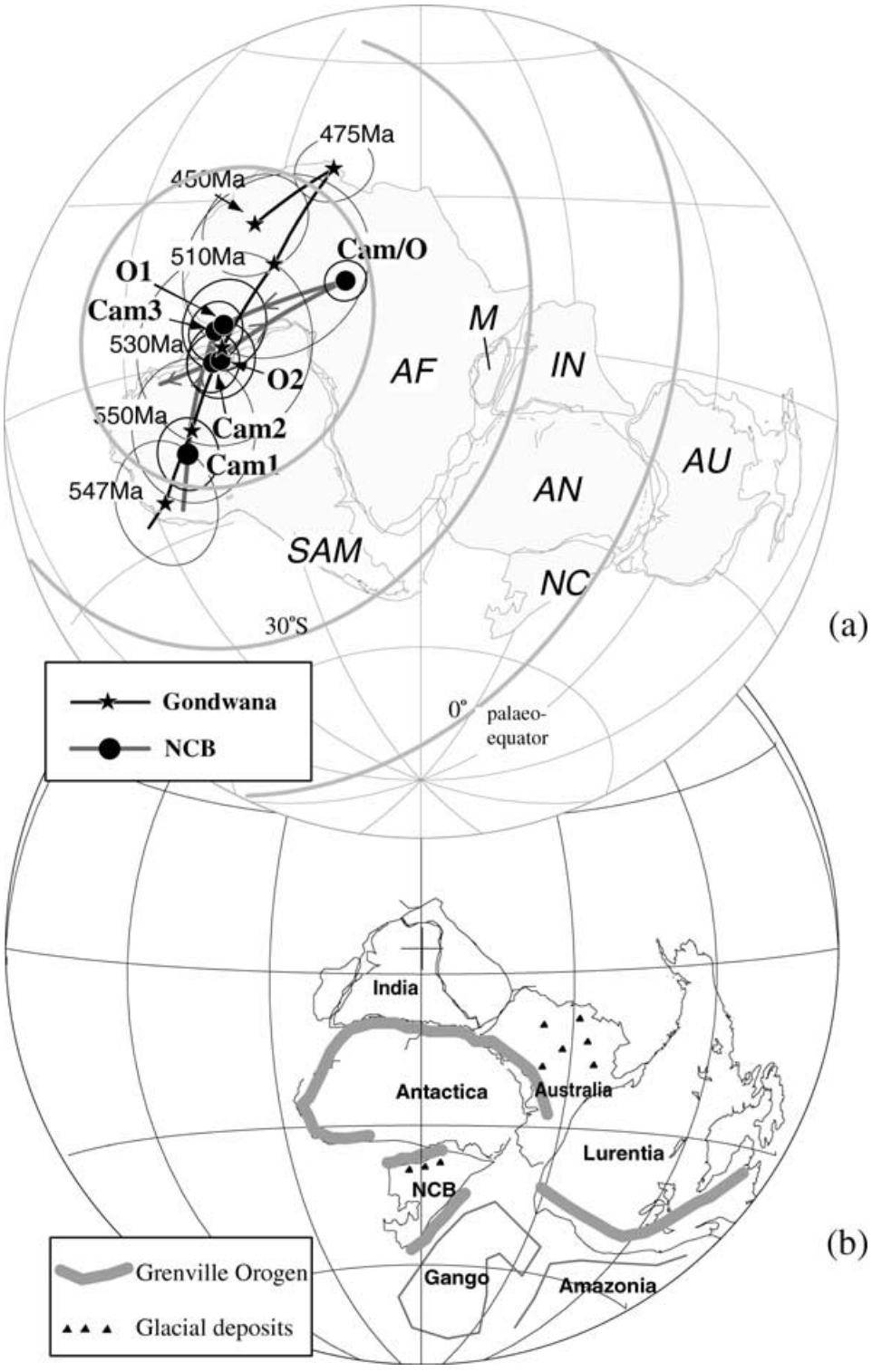


Figure 7. (a) Palaeomagnetically based palaeogeographic reconstruction between Gondwana and North China for the Cambrian and lowest Ordovician in African coordinates. This reconstruction is made by rotating the apparent south polar wander path of North China, which coincides with that of Gondwana, 146° clockwise around an Euler pole at 71.6°E and 5.8°N. Gondwanan reconstruction and APWP in African coordinates after de Wit *et al.* (1988) and Meert & Van der Voo (1997). South palaeopoles for North China: Middle Ordovician 147.7°E and 31.5°S ($\alpha 95 = 7.0°$, Average of poles from Zhao *et al.* 1992; Yang *et al.* 1997) Huang *et al.* 1999, Lower Ordovician 144.3°E and 37.4°S ($\alpha 95 = 8.5°$ Huang *et al.* 1999), Upper Cambrian 149.6°E and 31.7°S ($\alpha 95 = 5.4°$ Huang *et al.* 1999), Middle Cambrian 146.7°E 37.0°S ($\alpha 95 = 5.5°$ Huang *et al.* 1999), Lower Cambrian 161.9°E and 18.5°S ($\alpha 95 = 6.5°$ Huang *et al.* 1999). AF—Africa, AN—Antartica, AU—Australia, IN—India, NC—North China, M—Madagasca, SAM—South America. Palaeolatitudinal zones derived from Late Cambrian pole of North China is also shown. (b) The reconstruction of the Proterozoic supercontinent Rodinia, to which Australia connected to the Western US acronymized as AUSWUS (e.g. Karlstrom *et al.* 1999), shows the northern Qinling fold belt (Kuanping complex) of the NCB may be the western continuation of the Grenvillian belt in the eastern margin of Laurentia. Laurentia rotated to Africa in present coordinate about Euler pole at 45°N, 129°E, rotation 162°.

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determined from Australia, North Europe and Siberia during the Cambrian–Ordovician and lowest Ordovician time.

These results resolve the early Palaeozoic magnetic polarity uncertainty for the NCB. The NCB was located in the southern hemisphere at a low palaeolatitude (∼17°) during the Cambrian–Ordovician and the lowest Ordovician. By comparing the APWP from the NCB and that of Gondwana, we suggest that the NCB formed a part of Gondwana during Cambrian and lowest Ordovician times, and started to break away from Gondwana in the lower Ordovician. The reconstruction proposed is contrary to some palaeomagnetic models where the NCB was quite separate from Gondwana in the late Proterozoic, and was attached to the 'Pacific' side of Antarctica.

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